# FROM A UNIFORM HÖLDER BOUND TO A VALID LIMITING EQUATION

An a priori estimate is useful only after its topology and constants are specified. This lecture proves the compactness step and then separates the two further tasks: preserving the equation's domain and passing its residual.

## 1. The conditional target

We assume multivariable calculus, compactness of bounded real sequences and smooth partitions of unity. The compactness, interpolation and admissibility arguments below are developed explicitly.

**Definition 1.1.** Fix a compact smooth $d$-manifold $M$ without boundary, a smooth background metric, a finite smooth atlas $(U_a, \phi_a)$ and a smooth partition of unity $(\chi_a)$ with $\operatorname{supp}\chi_a \Subset U_a$. For $0 < \gamma < 1$, extend $v_a = (\chi_a u)\circ\phi_a^{-1}$ by zero to $\mathbb{R}^d$ and set

$$\|u\|_{2,\gamma} = \sum_a \left( \sum_{|\nu|\le 2} \|D^\nu v_a\|_\infty + \sum_{|\nu|=2} [D^\nu v_a]_\gamma \right), \qquad [w]_\gamma = \sup_{x\neq y} \frac{|w(x)-w(y)|}{|x-y|^\gamma}.$$

Here $\nu$ is a multi-index, and $C^{2,\gamma}(M)$ consists of the $C^2$ functions for which this norm is finite. All these choices stay fixed throughout a sequence.

**Theorem 1.2.** *Fix the compact smooth manifold and norms just defined, integers $d, r \ge 1$, and $0 < \beta < \alpha < 1$. Let*

$$A : C^{2,\beta}(M) \longrightarrow C^0(M; \operatorname{Sym}_r(\mathbb{R}))$$

*be continuous for the uniform operator norm in the target. Define*

$$\mathcal{U} = \{u \in C^{2,\beta}(M) : A(u)(x) > 0 \text{ for every } x \in M\}, \qquad \mathcal{F} : [0,1]\times\mathcal{U} \longrightarrow C^0(M),$$

*and assume $\mathcal{F}$ is jointly continuous. Suppose $t_j \to t_*$ and $u_j \in C^{2,\alpha}(M)\cap\mathcal{U}$ satisfy, with fixed $C < \infty$ and $\delta > 0$,*

$$\mathcal{F}(t_j, u_j) = 0, \qquad \|u_j\|_{2,\alpha} \le C, \qquad A(u_j)(x) \ge \delta I_r \quad \text{for every } j, x.$$

*Then some subsequence converges in $C^{2,\beta}(M)$ to $u_* \in C^{2,\alpha}(M)$, with*

$$\|u_*\|_{2,\alpha} \le C, \quad A(u_*)(x) \ge \delta I_r \quad (x \in M), \quad u_* \in \mathcal{U}, \quad \mathcal{F}(t_*, u_*) = 0.$$

*This is conditional on the displayed bounds and continuity; it asserts neither existence of such solutions nor higher regularity or convergence in $C^{2,\alpha}$.*

The estimate and the positive constant $\delta$ are assumptions. A specific PDE must supply them, uniformly in $j$, from its own coefficients, geometry and normalization; this theorem does not do so.

The proof proceeds from fixed coordinate pieces to a single limit, then to $C^{2,\beta}$ convergence. Only after the margin has placed the limit in $\mathcal{U}$ do we invoke continuity of $\mathcal{F}$.

## 2. Constructing the convergent subsequence

Write $v_{a,j} = (\chi_a u_j)\circ\phi_a^{-1}$, extended by zero. The chart supports are fixed compact sets. The assumed norm bound gives

$$\|D^\nu v_{a,j}\|_\infty \le C \quad (|\nu| \le 2), \qquad [D^\nu v_{a,j}]_\alpha \le C \quad (|\nu| = 2).$$

The second derivatives have a common Hölder modulus. Lower derivatives have a common Lipschitz modulus by the next-derivative bound and integration along a segment in $\mathbb{R}^d$.

**A common subsequence.** For each of the finitely many derivative families, choose a countable dense set in its compact support. Enumerate all scalar evaluations and successively extract subsequences on which the next evaluation converges.

Take the diagonal of these nested subsequences. It converges at every listed point. To promote this to uniform convergence, choose a finite net from the dense set using the common modulus of continuity.

For any two late terms $w_j, w_k$ of one derivative family and a net point $s$ close to $x$, use

$$|w_j(x) - w_k(x)| \le |w_j(x) - w_j(s)| + |w_j(s) - w_k(s)| + |w_k(s) - w_k(x)|.$$

The first and third terms are small by the common modulus. The middle term is uniformly small on the finite net, so the subsequence is uniformly Cauchy. This is the scalar Arzelà–Ascoli argument of [1].

**The limits really are derivatives.** Uniform limits of separately named derivatives need compatibility. For $|\nu| \le 1$, keep the identity

$$D^\nu v_{a,j}(x + he_i) - D^\nu v_{a,j}(x) = \int_0^h D^{\nu+e_i} v_{a,j}(x + se_i)\, ds.$$

Uniform convergence on the segment permits passage under the integral. The limiting identity identifies first and second derivatives, so each local limit is $C^2$ with the derivatives just extracted.

Fixed smooth transition rules pass to the limit as well. The partition of unity glues these compatible pieces to a global function $u_*$. Compact support inside each chart makes zero extension harmless.

For any fixed pair $x \ne y$, passing to the limit in its Hölder quotient retains the $\alpha$ bound. Taking suprema and then the finite sum yields

$$u_* \in C^{2,\alpha}(M), \qquad \|u_*\|_{2,\alpha} \le C.$$

## 3. Why the exponent must be weakened

Uniform convergence of second derivatives does not by itself control their $\beta$ seminorms. For a bounded $\alpha$-Hölder function $w$, two available bounds can be combined:

$$|w(x) - w(y)| \le \min\{2\|w\|_\infty, [w]_\alpha |x - y|^\alpha\}.$$

For positive $a, b$ and $0 < \theta < 1$, $\min(a, b) \le a^{1-\theta} b^\theta$. Set $\theta = \beta/\alpha$ and divide by $|x - y|^\beta$:

$$[w]_\beta \le (2\|w\|_\infty)^{1-\beta/\alpha} [w]_\alpha^{\beta/\alpha}. \tag{3.1}$$

The zero cases follow directly. Apply this to each difference $D^\nu(v_{a,j} - v_{a,*})$ with $|\nu| = 2$: its sup norm tends to zero and its $\alpha$ seminorm is at most $2C$.

The strictly positive exponent $1-\beta/\alpha$ forces the right side of (3.1) to vanish. Combined with the lower derivatives, this proves

$$\|u_j - u_*\|_{2,\beta} \longrightarrow 0$$

along the extracted subsequence. The theorem's stronger regularity of the limit is a boundedness conclusion, not convergence in that stronger norm.

## 4. Passing the domain before the equation

Continuity of $A$ now supplies uniform matrix convergence. For every $x \in M$ and $v \in \mathbb{R}^r$,

$$v^T A(u_*)(x)v = \lim_j v^T A(u_j)(x)v \geq \delta|v|^2.$$

Thus $A(u_*) \geq \delta I_r > 0$, and $u_* \in \mathcal{U}$. The pair $(t_*, u_*)$ is therefore in the actual domain of $\mathcal{F}$, where joint continuity gives

$$\mathcal{F}(t_*, u_*) = \lim_j \mathcal{F}(t_j, u_j) = 0 \quad \text{in } C^0(M).$$

This completes Theorem 1.2. If the sequence also has mean zero, that normalization passes because

$$\left| \int_M (u_j - u_*)\, dV_g \right| \leq \mathrm{Vol}_g(M)\|u_j - u_*\|_\infty \longrightarrow 0.$$

**A concrete continuity check.** On a flat torus take a fixed continuous $h$ and

$$\mathcal{F}(t, u) = \mathrm{tr}(D^2 u) + u^3 - th.$$

The second derivatives converge uniformly. On the common bounded range $|u_j|, |u_*| \leq R$, the nonlinear term satisfies

$$\|u_j^3 - u_*^3\|_\infty \leq 3R^2\|u_j - u_*\|_\infty.$$

The parameter term is bounded by $|t_j - t_*|\|h\|_\infty$. These three estimates verify the claimed $C^0$ continuity for this example; they provide no existence theorem or a priori solution bound.

## 5. Localization and failure models

On an open subset of $\mathbb{R}^d$, estimates on every relatively compact open set give a local conclusion. Choose an exhaustion $K_\ell$ with $K_\ell \Subset \mathrm{int}\, K_{\ell+1}$ and insert fixed cutoffs.

Extract on $K_1$, then on $K_2$ inside that subsequence, and continue. The diagonal converges on every fixed $K_\ell$; limits agree on overlaps. The bounds may depend on $\ell$, so the conclusion is only $C^{2,\beta}_{\mathrm{loc}}$.

For a nonzero compactly supported smooth function $\eta$ on $\mathbb{R}$, $u_j(x) = \eta(x-j)$ eventually vanishes on every fixed compact set, but

$$\|u_j\|_\infty = \|\eta\|_\infty > 0.$$

Thus local extraction does not imply global uniform convergence. It also supplies no boundary convergence on a noncompact or open domain.

**Losing the positive margin.** Let $A(u) = uI_r$, let $\mathcal{U}$ be the positive functions, and set $\mathcal{F}(t,u) = u - t$. The constant functions

$$t_j = u_j = 1/j$$

have uniform Hölder bounds and solve the equation, but their limit is zero and leaves $\mathcal{U}$. Individual strict inequalities cannot replace a uniform $\delta > 0$.

**Losing the exponent gap.** On $\mathbb{R}/(2\pi\mathbb{Z})$, let $u_n(x) = n^{-2-\alpha}\cos(nx)$. Its second derivative is $-n^{-\alpha}\cos(nx)$ and converges uniformly to zero. The bound $|\cos a - \cos b| \le \min\{2, |a-b|\}$ gives uniform $\alpha$ seminorms.

At $x = 0$ and $y = \pi/n$, however, the second-derivative quotient is

$$\frac{|u_n''(0) - u_n''(\pi/n)|}{(\pi/n)^\alpha} = \frac{2}{\pi^\alpha}.$$

It cannot converge to zero in $C^{2,\alpha}$. Since every possible uniform limit is zero, no subsequence can repair that failure.

**Weak convergence is a different input.** Work in the real space $L^2(0, 2\pi; \mathbb{R})$, with inner product $\langle f, g\rangle = \int_0^{2\pi} fg\,dx$ and norm $\|g\|_2^2 = \langle g, g\rangle$.

By definition, $f_n \rightharpoonup 0$ weakly in this space means $\int_0^{2\pi} f_n g\,dx \to 0$ for every $g \in L^2(0, 2\pi; \mathbb{R})$.

Set $e_n(x) = \sin(nx)/\sqrt{\pi}$ for integers $n \ge 1$. Integrating the product-to-sum identity

$$2\sin(nx)\sin(mx) = \cos((n-m)x) - \cos((n+m)x)$$

gives $\langle e_n, e_m\rangle = 0$ for $n \ne m$ and 1 for $n = m$. Thus the sequence is orthonormal.

Fix any $g$ in this real $L^2$ space and put $a_n = \langle g, e_n\rangle$. Expanding a finite squared norm using orthonormality gives

$$0 \le \left\| g - \sum_{n=1}^{N} a_n e_n \right\|_2^2 = \|g\|_2^2 - \sum_{n=1}^{N} |a_n|^2.$$

Hence the partial sums of $\sum |a_n|^2$ are bounded, so $a_n \to 0$. For this arbitrary $g$, $\int_0^{2\pi} \sin(nx) g(x)\,dx = \sqrt{\pi}\, a_n \to 0$, proving $\sin(nx) \rightharpoonup 0$.

The squares also lie in $L^2$. Testing against $1 \in L^2$ gives

$$\int_0^{2\pi} \sin^2(nx)\,dx = \pi.$$

Thus the squares do not converge weakly to zero. The nonlinear passage in the theorem instead uses strong convergence in a topology for which continuity was actually verified.

## References


[1] Joel H. Shapiro. The arzela-ascoli theorem. Online source, 1999. unpublished; controlling version: Math 829, Spring 1999.